

University
of Padua

Faculty of Data Science

SUBSPACE IDENTIFICATION METHODS FOR EXTRACTING BEHAVIORALLY RELEVANT NEURAL DYNAMICS

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INTRODUCTION

Introduction

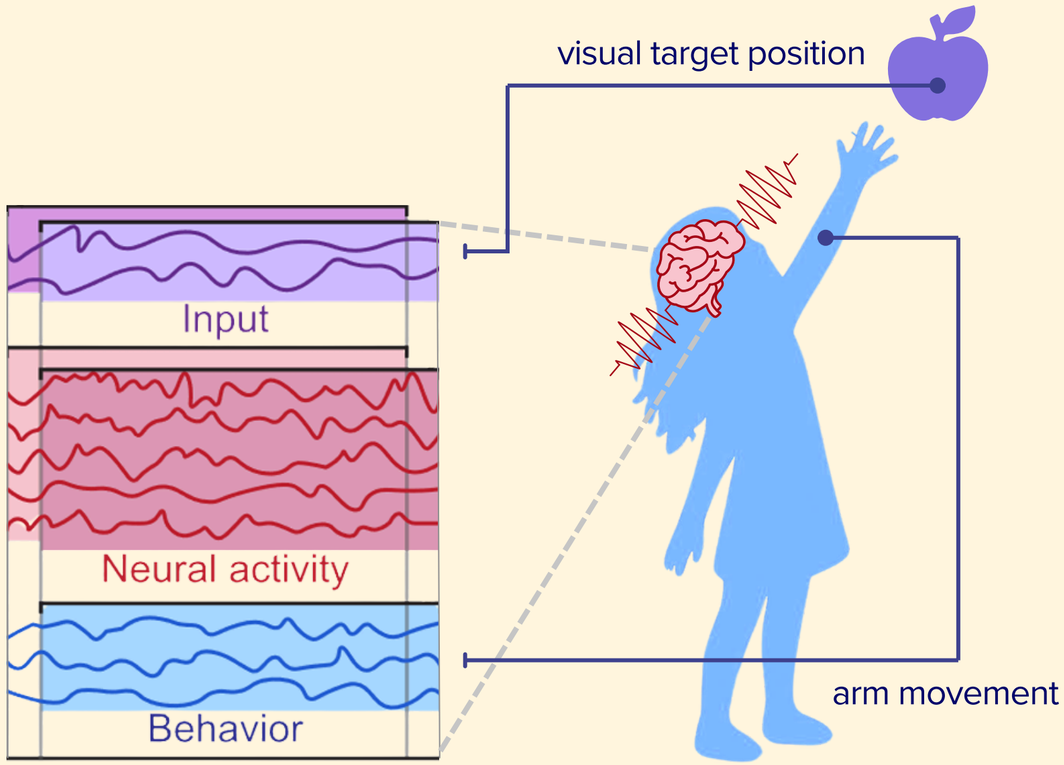
Understanding neural signals is important:

➜ *brain theories*

➜ practical applications: *brain-machine interfaces*

However, **brain activity** is the result of **different contributions**:

- ***input-driven*** activity
- ***behavioral*** neural activity
- ***other*** neural activity



Example: a human reaching for an apple

Modeling Neural Data

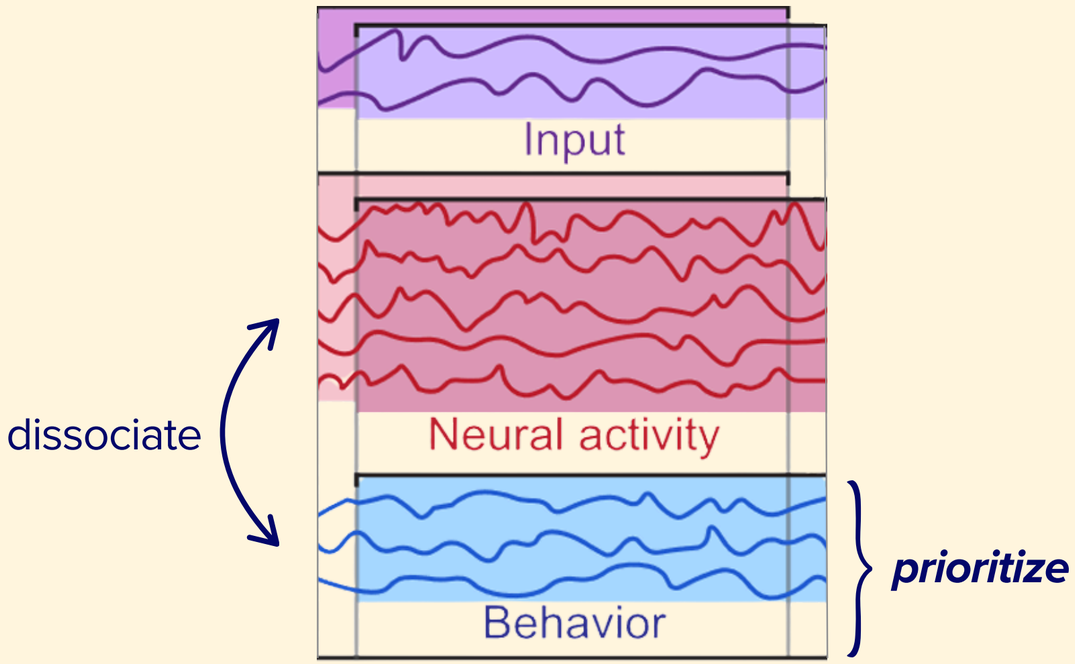
1

Prioritize Behavior *(Objective 1)*

Dissociate and ***prioritize behaviorally relevant*** dynamics from other dynamics

Why?

- Most neural variance is behaviorally irrelevant
- No prioritization → model captures dominant (not behavior-predictive) dynamics



Modeling Neural Data

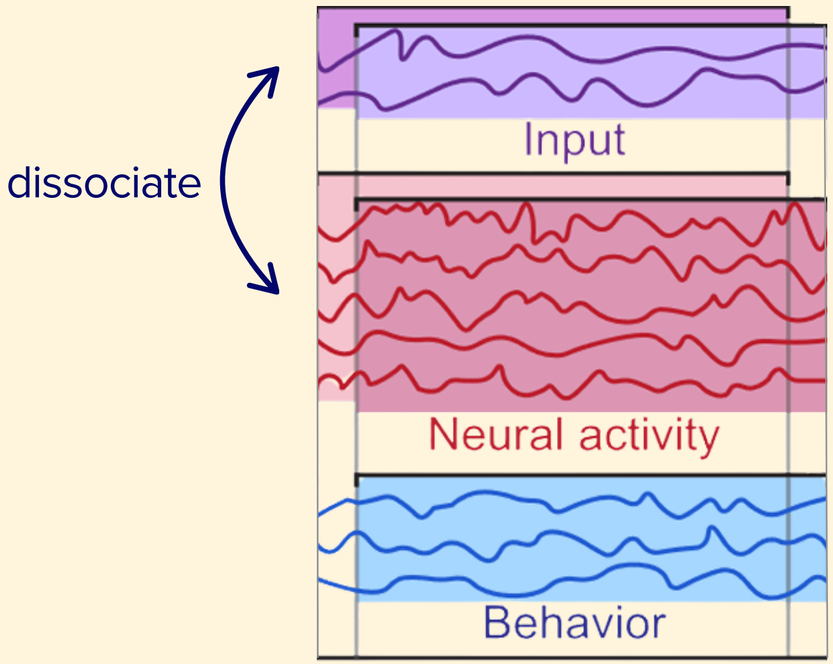
2

Dissociate Input *(Objective 2)*

Dissociate intrinsic neural dynamics ***from input-driven*** dynamics

Why?

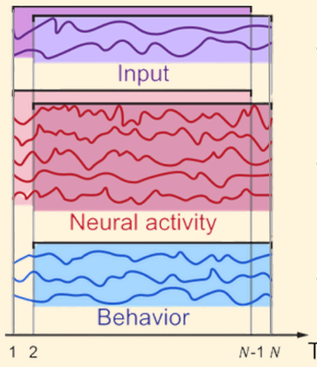
- Inputs carry their own temporal structure
- No dissociation ➜ input dynamics are mistaken for intrinsic ones



FRAMEWORK AND MODELS

General Framework

Measurements



- **u_k input**
- **y_k neural activity**
- **z_k behavior**

State-space model

The measurements evolution over time is modeled via a **dynamical system**

Non-preferential methods (NDM, INDM)

Preferential methods (PSID, IPSID)

$$\begin{cases} x_{k+1} = Ax_k + Bu_k + w_k, \\ y_k = C_y x_k + D_y u_k + v_k, \\ z_k = C_z x_k + D_z u_k + \epsilon_k \end{cases}$$

x_k latent state → low dimensional vector encoding neural dynamics

Latent state extraction
+
System matrices estimation

Predict behavior

Given new neural recordings, **behavior can be predicted** without directly observing it

PSID Formulation

Preferential Subspace Identification (PSID) models the evolution of behavior and neural activity with the following linear system:

$$\begin{cases} x_{k+1} = Ax_k + w_k, \\ y_k = C_y x_k + v_k, \\ z_k = C_z x_k + \epsilon_k \end{cases} \quad , \quad x_k = \begin{bmatrix} x_k^{(1)} \\ x_k^{(2)} \end{bmatrix}$$

- **y_k neural activity**
- **z_k behavior**
- **x_k latent states**:
  behaviorally ***relevant*** (top n_1 rows)
  behaviorally ***irrelevant*** (bottom n_2)

Key structural constraint to prioritize behaviorally relevant dynamics:

$$C_z = [C_{z_1} \; 0]$$

➜ reaches *Objective 1:* Prioritize Behavior

IPSID Formulation

Input Preferential Subspace Identification (IPSID) extends PSID to input-driven settings:

$$\begin{cases} x_{k+1} = Ax_k + Bu_k + w_k, \\ y_k = C_y x_k + D_y u_k + v_k, \\ z_k = C_z x_k + D_z u_k + \epsilon_k \end{cases} \quad , \quad x_k = \begin{bmatrix} x_k^{(1)} \\ x_k^{(2)} \end{bmatrix}$$

- **y_k neural activity**
- **z_k behavior**
- **u_k input**
- **x_k latent states**:
  behaviorally ***relevant*** (top n_1 rows)
  behaviorally ***irrelevant*** (bottom n_2)

Same structural constraint to prioritize behaviorally relevant dynamics:

$$C_z = [C_{z_1} \; 0]$$

➜ reaches *Objective 1:* Prioritize Behavior
Objective 2: Dissociate Input

Algorithm Outline

Inputs:

- neural activity
- behavior
- (input data)

Stage 1
Behaviorally relevant states

Extract $\mathbf{X_i^{(1)}}$ **from behavior** via orthogonal projection.

Stage 2
Behaviorally irrelevant states

Extract $\mathbf{X_i^{(2)}}$ **from neural activity** via orthogonal projection.

Final Stage
Parameter identification

Identify the **system parameters** via least-squares regression.

Outputs:

- latent states
- system matrices

Stage 1

row space of $\mathbf{X_i^{(1)}}$ = row space of the **component of $\mathbf{Z_f}$** that lies in the linear **space spanned by** the rows of $\mathbf{Y_p}$

$$\hat{X}_i^{(1)} = \Gamma^{(1)} \left(Z_f / Y_p \right)$$

$$Y_f' = Y_f - Y_f / \hat{X}_i^{(1)}$$

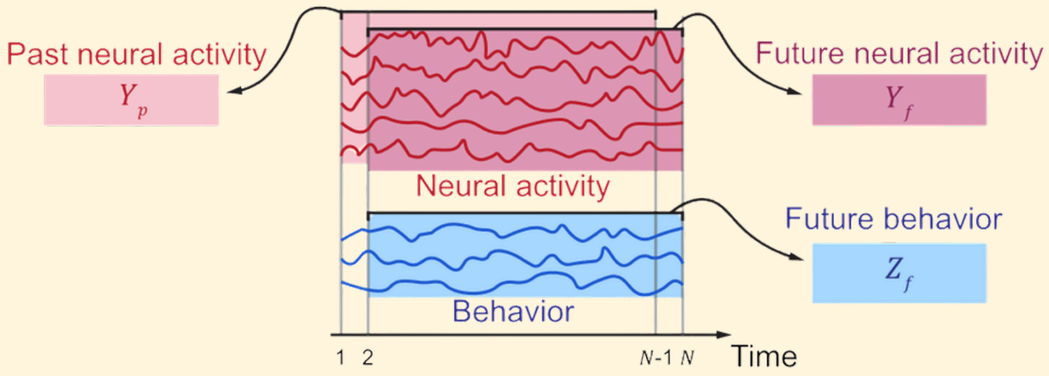
neural activity predicted by $X_i^{(1)}$

Stage 2

row space of $X_i^{(2)}$ = row space of the **component of $\mathbf{Y_f'}$** that lies in the linear **space spanned by** the rows of $\mathbf{Y_p}$

$$\hat{X}_i^{(2)} = \Gamma^{(2)} \left(Y_f' / Y_p \right)$$

Inputs



Final Stage

$$\begin{cases} x_{k+1} = Ax_k + w_k, \\ y_k = C_y x_k + v_k, \\ z_k = C_z x_k + \epsilon_k \end{cases}$$

Parameter identification via least-squares regression

NUMERICAL SIMULATIONS

Numerical Simulations

40 random state-space models are generated → **synthetic neural activity, behavior,** and **input** time series
(20 PSID, 20 IPSID)

1
EIGENVALUE RECOVERY

- Extract the *eigenvalues* of the estimated *behaviorally relevant submatrix* A_{11} and compare them to the true ones

- **Goal**: assess accuracy of ***behaviorally relevant dynamics recovery***
- **Metric**: normalized eigenvalue error E

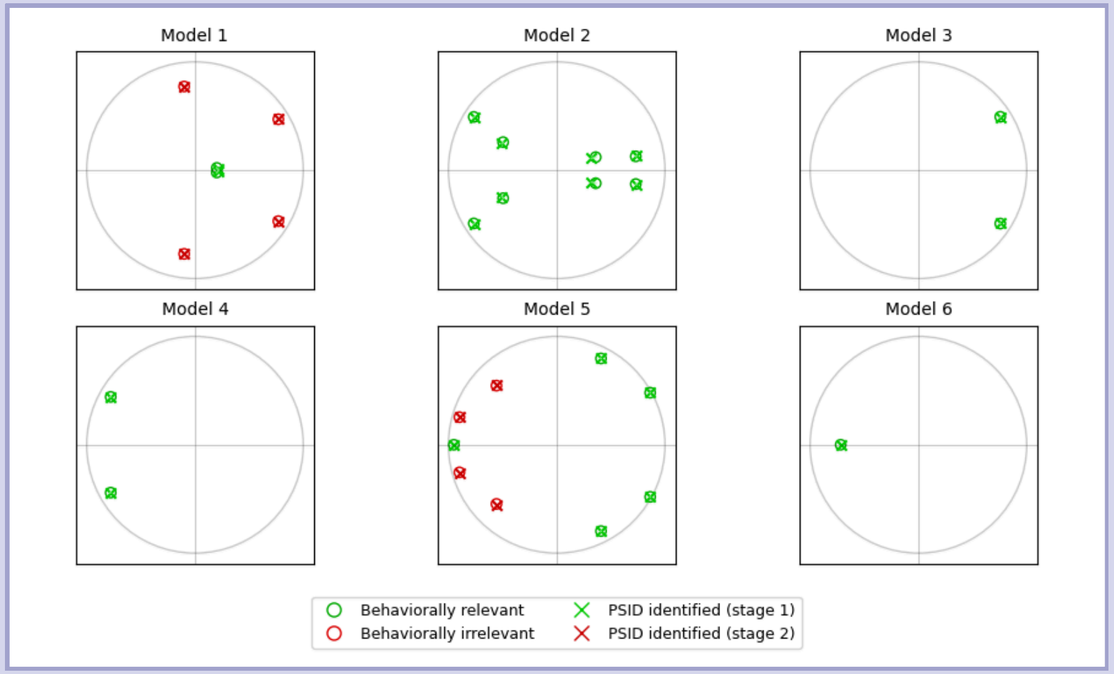
2
BEHAVIOR DECODING

- Extracts *latent states* from test neural activity
- Predict behavior via the linear observation equation

- **Goal**: assess accuracy of ***behavior decoding***
- **Metric**: R^2

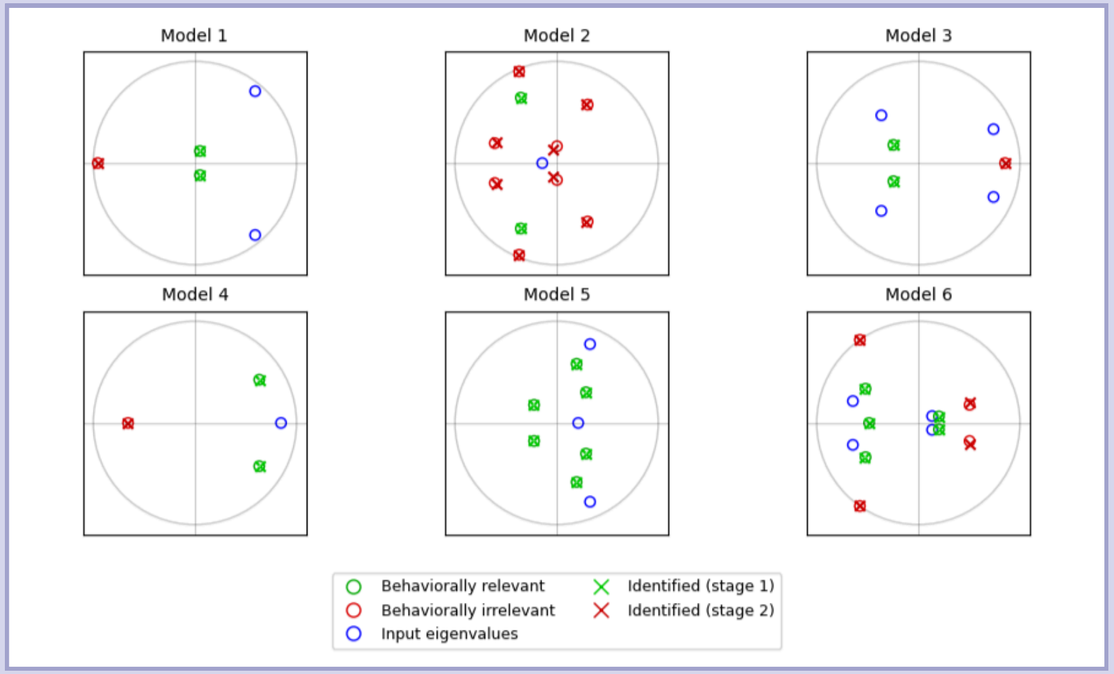
Results – *Eigenvalue Recovery*

PSID



- **Correctly recovered behaviorally relevant dynamics**
- Small normalized eigenvalue error:
  range → 0.0014 – 0.1277
  median error → **0.011**

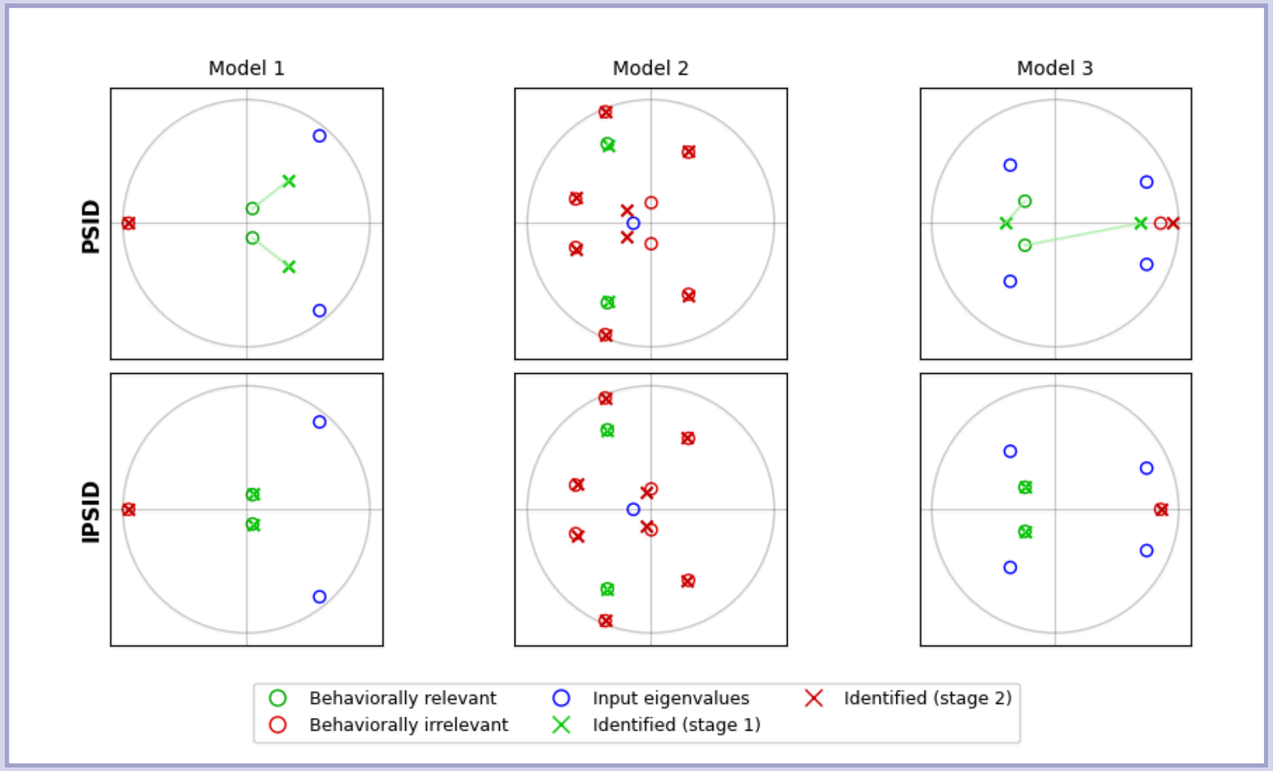
IPSID



- **Correctly recovered behaviorally relevant dynamics**
- Small normalized eigenvalue error:
  range → 0.0001 – 0.0997
  median error → **0.003**
- **Input** eigenvalues are **not confused with intrinsic ones**

Results – *Eigenvalue Recovery*

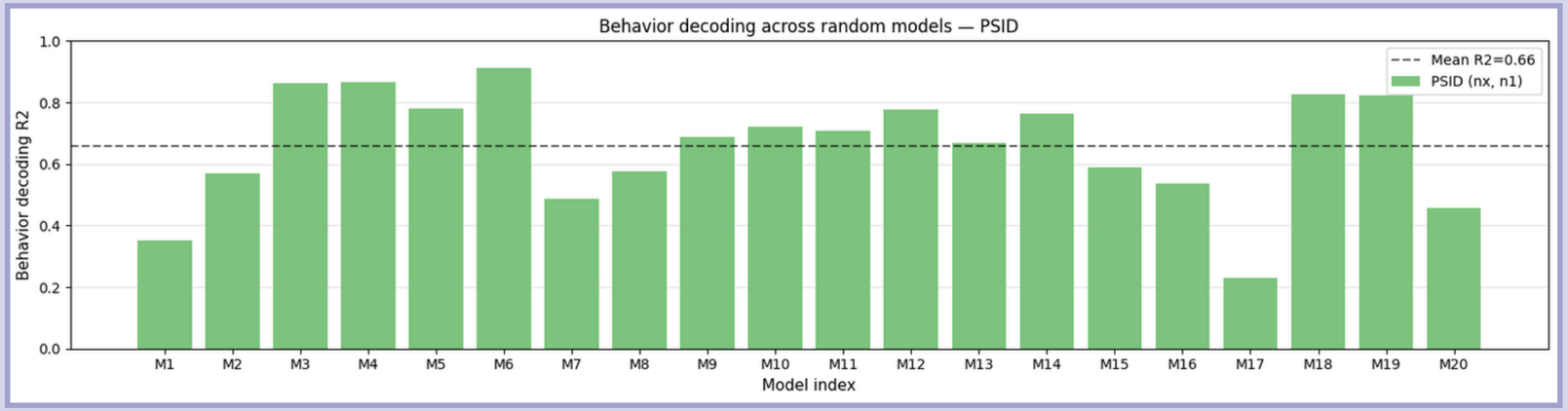
What happens if measured inputs are ignored?



- PSID identified eigenvalues are **displaced toward input eigenvalues** (E up to 2.905)
- IPSID recovers them accurately (E as low as 0.001)

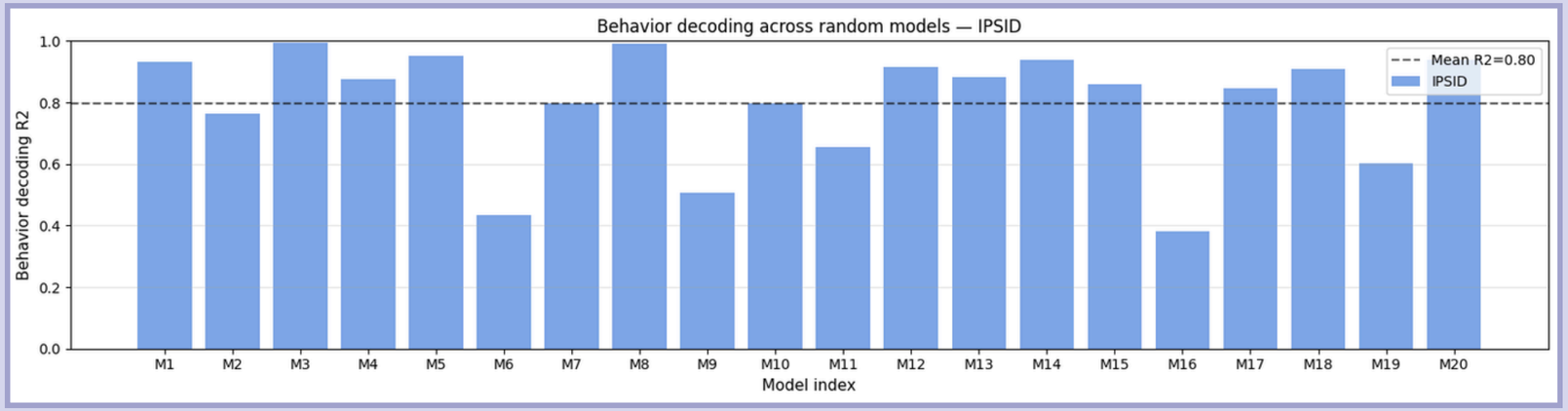
➔ Explicitly accounting for input is essential

Results – *Behavior Decoding*



PSID

- mean R^2 of **0.66**
  range: 0.231 – 0.911
  ➜ generally good behavior decoding



IPSID

- mean R^2 of **0.80**
  range: 0.382 – 0.993
  ➜ accurate behavior decoding

CONCLUSIONS

Conclusions

Summary of Contributions

- PSID and IPSID efficiently **prioritize behaviorally relevant dynamics**
- IPSID also efficiently **dissociates input** dynamics

Improved understanding of neural data

Future Directions

→ Validate on real neural data

→ Explore non linear state-space systems

→ Compare with neural network models

→ Adapt to non-stationary dynamics



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THANK YOU

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Appendix
References

[1] O. G. Sani, H. Abbaspourazad, Y. T. Wong, B. Pesaran, and M. M. Shanechi, ***"Modeling behaviorally relevant neural dynamics enabled by preferential subspace identification"***, Nature Neuroscience, vol. 24, no. 1, pp. 140–149, 2021

[2] P. Vahidi, O. G. Sani, and M. M. Shanechi, ***"Modeling and dissociation of intrinsic and input-driven neural population dynamics underlying behavior"***, Proceedings of the National Academy of Sciences, vol. 121, no. 7, p. e2212887121, 2024

[3] P. Van Overschee and B. De Moor, ***Subspace Identification for Linear Systems.*** Springer, 1996

Appendix
Results

Summary of PSID (left) and IPSID (right) simulation results across randomly generated state-space models. Each method was evaluated on an independent set of 20 models, characterized by their total latent state dimension nx and behaviorally relevant state dimension n1. For each model, the normalized eigenvalue identification error Eλ and the behavior decoding R2 are reported.

PSID

Model	n_x	n_1	$\mathcal{E}_\lambda$	R^2
M1	6	2	0.0975	0.353
M2	8	8	0.0273	0.569
M3	2	2	0.0031	0.861
M4	2	2	0.0047	0.865
M5	9	5	0.0025	0.779
M6	1	1	0.0068	0.911
M7	6	4	0.0218	0.487
M8	7	7	0.0177	0.577
M9	3	3	0.0072	0.689
M10	10	4	0.0031	0.719
M11	3	1	0.0153	0.706
M12	8	4	0.0019	0.776
M13	10	10	0.0390	0.669
M14	9	8	0.0273	0.763
M15	7	3	0.0062	0.589
M16	8	8	0.0163	0.535
M17	9	4	0.1277	0.231
M18	2	2	0.0059	0.825
M19	3	3	0.0014	0.825
M20	4	4	0.0305	0.457

IPSID

Model	n_x	n_1	$\mathcal{E}_\lambda$	R^2
M1	3	2	0.0314	0.931
M2	10	2	0.0019	0.762
M3	3	2	0.0011	0.993
M4	3	2	0.0061	0.875
M5	6	6	0.0033	0.952
M6	9	5	0.0124	0.433
M7	4	4	0.0010	0.797
M8	9	7	0.0009	0.990
M9	10	2	0.0191	0.505
M10	3	2	0.0019	0.797
M11	3	3	0.0142	0.655
M12	5	4	0.0034	0.914
M13	10	2	0.0023	0.881
M14	9	1	0.0005	0.939
M15	4	2	0.0014	0.859
M16	9	4	0.5735	0.382
M17	5	5	0.0167	0.846
M18	2	2	0.0022	0.908
M19	8	8	0.1672	0.602
M20	10	4	0.0019	0.937